

Note on Some Pseudo-Elliptic Integrals

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Consider a differential of the form

$$f(x) \frac{dx}{\sqrt{Ax^2 + Bx + C}}, \quad (1)$$

where $f(x)$ denotes an arbitrary rational function and $Ax^2 + Bx + C$ is a second-degree trinomial that is not a perfect square, which may, as a special case, reduce to a linear expression. Let us pose the substitution

$$x = \frac{at^2 + bt + c}{a't^2 + b't + c'} = \varphi(t); \quad (2)$$

the numerator and the denominator of $\varphi(t)$ are assumed to be coprime, and at least one is of the second degree. By this substitution, the differential (1) becomes

$$\mathfrak{F}(t) \frac{dt}{\sqrt{R(t)}} \quad (3)$$

where

$$\begin{aligned} \mathfrak{F}(t) &= f \left(\frac{at^2 + bt + c}{a't^2 + b't + c'} \right) \frac{Lt^2 - 2Nt + M}{(a't^2 + b't + c')^2} \\ L &= ab' - ba', \quad M = bc' - cb', \quad N = ca' - ac', \\ R(t) &= A(at^2 + bt + c)^2 + B(at^2 + bt + c)(a't^2 + b't + c') + C(a't^2 + b't + c')^2. \end{aligned}$$

The two roots t_1, t_2 of the equation $\varphi(t) = \lambda$ satisfy, for any λ , the involution relation

$$Lt_1t_2 - N(t_1 + t_2) + M = 0,$$

obtained by eliminating λ between the two equations

$$\begin{aligned} a(t_1 + t_2) + b - \lambda[a'(t_1 + t_2) + b'] &= 0, \\ at_1t_2 - c - \lambda[a't_1t_2 - c'] &= 0. \end{aligned}$$

It follows that if, in $\varphi(t)$, we make the linear substitution

$$Ltz - N(t + z) + M = 0, \quad \text{or} \quad t = \frac{Nz - M}{Lz - N}, \quad (4)$$

we will have identically $\varphi(t) = \varphi(z)$. Indeed, one can verify without any difficulty the identities below:

$$\begin{aligned} at^2 + bt + c &= (N^2 - LM) \frac{az^2 + bz + c}{(Lz - N)^2}; \\ a't^2 + b't + c' &= (N^2 - LM) \frac{a'z^2 + b'z + c'}{(Lz - N)^2}; \\ Lt^2 - 2Nt + M &= (LM - N^2) \frac{Lz^2 - 2Nz + M}{(Lz - N)^2}. \end{aligned}$$

We therefore have

$$\begin{aligned} \frac{dt}{\sqrt{R(t)}} &= \frac{(LM - N^2)dz}{\sqrt{(N^2 - LM)^2 R(z)}} = \frac{dz}{\sqrt{R(z)}}, \\ f\left(\frac{at^2 + bt + c}{a't^2 + b't + c'}\right) &= f\left(\frac{az^2 + bz + c}{a'z^2 + b'z + c'}\right); \\ \frac{Lt^2 - 2Nt + M}{(a't^2 + b't + c')^2} &= -\frac{Lz^2 - 2Nz + M}{(a'z^2 + b'z + c')^2}. \end{aligned}$$

It follows that

$$\mathfrak{F}(t) = f\left(\frac{at^2 + bt + c}{a't^2 + b't + c'}\right) \frac{Lt^2 - 2Nt + M}{(a't^2 + b't + c')^2} = -f\left(\frac{az^2 + bz + c}{a'z^2 + b'z + c'}\right) \frac{Lz^2 - 2Nz + M}{(a'z^2 + b'z + c')^2} = -\mathfrak{F}(z).$$

The differential (3) therefore enjoys the following property:

There exists an involutive linear substitution $t = \frac{Nz-M}{Lz-N}$, which permutes the roots of the equation $R(t) = 0$ in pairs, such that we identically have $\mathfrak{F}\left(\frac{Nz-M}{Lz-N}\right) = -\mathfrak{F}(z)$.

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Conversely, let $R(t)$ be a fourth-degree polynomial having four distinct roots a, b, c, d , and let

$$Lt_1t_2 - N(t_1 + t_2) + M = 0$$

be an involutive linear substitution which permutes the roots a, b, c, d in pairs. If a rational function $\mathfrak{F}(t)$ is such that we identically have

$$\mathfrak{F}(t_1) + \mathfrak{F}(t_2) = 0,$$

the integral

$$\int \mathfrak{F}(t) \frac{dt}{\sqrt{R(t)}}$$

is a pseudo-elliptic integral. Let us indeed denote by α and β the double points of the preceding involution; the relation between t_1 and t_2 can be written

$$\frac{t_1 - \alpha}{t_1 - \beta} + \frac{t_2 - \alpha}{t_2 - \beta} = 0.$$

Let's make the change of variable,

$$\frac{t - \alpha}{t - \beta} = u;$$

the differential $\frac{dt}{\sqrt{R(t)}}$ is changed into a differential of the form $\frac{du}{\sqrt{R_1(u)}}$, where the polynomial $R_1(u)$ has only terms of even degree; because the relation between the values u_1, u_2 of u , which

correspond to the values t_1, t_2 of t , is precisely $u_1 + u_2 = 0$. Similarly, the rational function $\mathfrak{F}(t)$ is changed into a certain rational function $F(u)$, and the relation

$$\mathfrak{F}(t_1) + \mathfrak{F}(t_2) = 0$$

becomes

$$F(u_1) + F(u_2) = 0, \quad \text{that is} \quad F(u) + F(-u) = 0.$$

It follows from this that $F(u)$ will be of the form $uF_1(u^2)$, and it will suffice to set $u^2 = x$ to reduce the differential

$$\frac{uF_1(u^2)du}{\sqrt{R_1(u)}}$$

to a differential of the form (1). We can therefore state the following theorem:

Theorem 1. Let $\left[t, \frac{Nt-M}{Lt-N}\right]$ be a substitution of period 2 permuting the four roots a, b, c, d of the equation $R(t) = 0$ in pairs, and let $\mathfrak{F}(t)$ be a rational function such that one identically has

$$\mathfrak{F}(t) + \mathfrak{F}\left(\frac{Nt-M}{Lt-N}\right) = 0;$$

the integral

$$\int \mathfrak{F}(t) \frac{dt}{\sqrt{R(t)}}$$

is a pseudo-elliptic integral.

This result was obtained in another way by M. Raffy (see Bulletin, t. XII, p. 51); the above process furthermore gives the means of effecting the reduction. The preceding statement still applies when $R(t)$ is of the third degree, provided one regards one of the four roots a, b, c, d as infinite.

Definition 1 (Involution). An **involution** is a function S that is its own inverse; that is, $S(S(t)) = t$ for all t in its domain. In the context of this theorem, we consider an involution that is a non-trivial *linear fractional transformation*, having the form:

$$S(t) = \frac{at+b}{ct+d}, \quad ad-bc \neq 0$$

Such a transformation always has two fixed points (which may coincide or include ∞), given by the solutions to $S(t) = t$.

Theorem (thm. 1. Goursat, 1887). Let $R(t)$ be a quartic polynomial with distinct roots, and let $S(t)$ be a non-trivial involution that permutes the roots of $R(t)$. If the rational function $\mathfrak{F}(t)$ satisfies the anti-invariance condition $\mathfrak{F}(t) + \mathfrak{F}(S(t)) = 0$, then the integral

$$I = \int \frac{\mathfrak{F}(t)}{\sqrt{R(t)}} dt$$

is elementary.

Proof. Let the two fixed points of the involution $S(t)$ be α and β . We introduce the change of variable

$$u = \frac{t-\alpha}{t-\beta}$$

Under this transformation, the involution S on t corresponds to the simple reflection $u \mapsto -u$. Let us examine the effect of this substitution on the integrand.

The rational function $\mathfrak{F}(t)$ transforms into a new rational function $F(u) = \mathfrak{F}(t(u))$. The condition $\mathfrak{F}(t) + \mathfrak{F}(S(t)) = 0$ becomes $F(u) + F(-u) = 0$, which implies that $F(u)$ is an odd function of u .

The polynomial $R(t)$ transforms into a new expression $R_1(u)$ under the radical. Since S permutes the roots of $R(t)$, the mapping $u \mapsto -u$ must permute the roots of $R_1(u)$. A polynomial whose roots occur in opposite pairs must be an even function. Thus, $R_1(u)$ is a polynomial in u^2 , which we denote by $R_2(u^2)$.

The differential form $\frac{dt}{\sqrt{R(t)}}$ can likewise be shown to transform into an even function of u . The integral therefore becomes

$$I = \int F(u) \cdot (\text{even function of } u) du = \int (\text{odd function of } u) du$$

The resulting integrand must have the structure $\frac{u \cdot H(u^2)}{\sqrt{R_2(u^2)}}$ for some rational function H .

We now apply a second substitution:

$$x = u^2 \implies dx = 2u du$$

The integral is reduced to:

$$I = \int \frac{H(x)}{2\sqrt{R_2(x)}} dx$$

Since $R(t)$ is a quartic, $R_1(u)$ is also a quartic, and therefore $R_2(x)$ must be a quadratic polynomial in x . The final integral is of the form $\int G(x)/\sqrt{ax^2 + bx + c} dx$ for a rational function $G(x)$. Such an integral is known to be elementary. $\square$

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The four roots a, b, c, d being arbitrary, there exist three linear substitutions of period 2, which permute these roots in pairs. For example, the substitution

$$\left\{ t, \frac{(ab - cd)t + (a + b)cd - (c + d)ab}{[(a + b) - (c + d)]t - (ab - cd)} \right\}$$

permutes the two roots a and b and the two roots c and d . The other two substitutions are deduced from this one, by successively taking any two of the roots for a and b , and the other two for c and d . Let us denote by S_1, S_2, S_3 these three substitutions; with the identical substitution $I(t) = t$, they form a group of four substitutions, and we have the relations

$$\begin{aligned} S_1^2 &= S_2^2 = S_3^2 = I, \\ S_1 S_2 &= S_2 S_1 = S_3, \\ S_1 S_3 &= S_3 S_1 = S_2, \\ S_2 S_3 &= S_3 S_2 = S_1. \end{aligned}$$

Let $S \left[t, \frac{Nt-M}{Lt-N} \right]$ be in general a linear substitution and $F(t)$ a rational function of t ; I will denote, for short, by $F(S)$ the rational function $F \left(\frac{Nt-M}{Lt-N} \right)$.

Given any polynomial of the fourth degree $R(t)$, we know from the above of three types of pseudo-elliptic integrals, relative to this polynomial,

$$\int f(t) \frac{dt}{\sqrt{R(t)}}, \quad \int \varphi(t) \frac{dt}{\sqrt{R(t)}}, \quad \int \psi(t) \frac{dt}{\sqrt{R(t)}},$$

the rational functions $f(t)$, $\varphi(t)$, $\psi(t)$ respectively satisfying the relations

$$f(S_1) + f = 0, \quad \varphi(S_2) + \varphi = 0, \quad \psi(S_3) + \psi = 0,$$

which are equivalent to these:

$$f(S_2) + f(S_3) = 0, \quad \varphi(S_1) + \varphi(S_3) = 0, \quad \psi(S_1) + \psi(S_2) = 0.$$

It is clear that by adding these three integrals, we still get a pseudo-elliptic integral

$$\int \mathfrak{F}(t) \frac{dt}{\sqrt{R(t)}};$$

where $\mathfrak{F}(t) = f(t) + \varphi(t) + \psi(t)$. The preceding relationships, which are satisfied by the functions f, φ, ψ , give us, for the function $\mathfrak{F}(t)$, the relation

$$\mathfrak{F} + \mathfrak{F}(S_1) + \mathfrak{F}(S_2) + \mathfrak{F}(S_3) = 0. \quad (5)$$

Conversely, let $\mathfrak{F}(t)$ be a rational function satisfying this condition (5). Let us set

$$f = \frac{\mathfrak{F} - \mathfrak{F}(S_1) + \mathfrak{F}(S_2) - \mathfrak{F}(S_3)}{4}, \quad \varphi = \frac{\mathfrak{F} + \mathfrak{F}(S_1) - \mathfrak{F}(S_2) - \mathfrak{F}(S_3)}{4}, \quad \dots$$

We can therefore state the following theorem:

Theorem 2. Let S_1, S_2, S_3 be the three linear period 2 substitutions that permute the four roots of the equation $R(t) = 0$ in pairs, and let $\mathfrak{F}(t)$ be a rational function such that one identically has

$$\mathfrak{F} + \mathfrak{F}(S_1) + \mathfrak{F}(S_2) + \mathfrak{F}(S_3) = 0;$$

the integral

$$\int \mathfrak{F}(t) \frac{dt}{\sqrt{R(t)}}$$

is a pseudo-elliptic integral.

Theorem (Goursat's Second Theorem). Let $R(t)$ be a quartic polynomial with distinct roots, and let S_1, S_2, S_3 be the three distinct involutions that permute the roots of $R(t)$. If a rational function $\mathfrak{F}(t)$ satisfies the summation condition

$$\mathfrak{F}(t) + \mathfrak{F}(S_1(t)) + \mathfrak{F}(S_2(t)) + \mathfrak{F}(S_3(t)) = 0$$

then the integral $I = \int \frac{\mathfrak{F}(t)}{\sqrt{R(t)}} dt$ is elementary.

Proof. The set of transformations $G = \{I, S_1, S_2, S_3\}$, where I is the identity map, forms a group isomorphic to the Klein four-group (V_4). The group properties include $S_1^2 = I$, $S_1 S_2 = S_3$, $S_2 S_1 = S_3$, etc.

The core of the proof is to decompose $\mathfrak{F}(t)$ into a sum of functions, each of which satisfies the simpler condition of Theorem 1. We can project $\mathfrak{F}(t)$ onto the subspaces corresponding to the non-trivial characters of V_4 .

Let us define three components of $\mathfrak{F}(t)$:

$$\begin{aligned} \mathfrak{F}_1(t) &= \frac{1}{4} (\mathfrak{F}(t) - \mathfrak{F}(S_1 t) - \mathfrak{F}(S_2 t) + \mathfrak{F}(S_3 t)) \\ \mathfrak{F}_2(t) &= \frac{1}{4} (\mathfrak{F}(t) - \mathfrak{F}(S_1 t) + \mathfrak{F}(S_2 t) - \mathfrak{F}(S_3 t)) \\ \mathfrak{F}_3(t) &= \frac{1}{4} (\mathfrak{F}(t) + \mathfrak{F}(S_1 t) - \mathfrak{F}(S_2 t) - \mathfrak{F}(S_3 t)) \end{aligned}$$

The summation condition given in the theorem statement, $\mathfrak{F} + \mathfrak{F}(S_1) + \mathfrak{F}(S_2) + \mathfrak{F}(S_3) = 0$, ensures that $\mathfrak{F}(t)$ is the sum of these three components:

$$\mathfrak{F}(t) = \mathfrak{F}_1(t) + \mathfrak{F}_2(t) + \mathfrak{F}_3(t)$$

Therefore, the integral can be split into three parts:

$$I = \int \frac{\mathfrak{F}_1(t)}{\sqrt{R(t)}} dt + \int \frac{\mathfrak{F}_2(t)}{\sqrt{R(t)}} dt + \int \frac{\mathfrak{F}_3(t)}{\sqrt{R(t)}} dt$$

The proof is complete if we can show that each of these three integrals is elementary. We do this by showing that each component $\mathfrak{F}_i$ satisfies the anti-invariance condition of Theorem 1 for at least one involution.

Let's check $\mathfrak{F}_1(t)$ for anti-invariance under S_1 :

$$\begin{aligned} 4(\mathfrak{F}_1(t) + \mathfrak{F}_1(S_1 t)) &= (\mathfrak{F} - \mathfrak{F}S_1 - \mathfrak{F}S_2 + \mathfrak{F}S_3) + (\mathfrak{F}S_1 - \mathfrak{F}S_1^2 - \mathfrak{F}S_2S_1 + \mathfrak{F}S_3S_1) \\ &= (\mathfrak{F} - \mathfrak{F}S_1 - \mathfrak{F}S_2 + \mathfrak{F}S_3) + (\mathfrak{F}S_1 - \mathfrak{F} - \mathfrak{F}S_3 + \mathfrak{F}S_2) \\ &= 0 \end{aligned}$$

Thus, $\mathfrak{F}_1(t) + \mathfrak{F}_1(S_1 t) = 0$. By Theorem 1, the integral $\int \frac{\mathfrak{F}_1(t)}{\sqrt{R(t)}} dt$ is elementary.

Similarly, we can show:

- $\mathfrak{F}_2(t)$ is anti-invariant under S_2 : $\mathfrak{F}_2(t) + \mathfrak{F}_2(S_2 t) = 0$.
- $\mathfrak{F}_3(t)$ is anti-invariant under S_3 : $\mathfrak{F}_3(t) + \mathfrak{F}_3(S_3 t) = 0$.

Therefore, the integrals for $\mathfrak{F}_2$ and $\mathfrak{F}_3$ are also elementary by Theorem 1.

Since the original integral I is a sum of three elementary integrals, it is itself elementary. $\square$

The theorem still applies when $R(t)$ is of the third degree, provided we look at one of the roots a, b, c, d as infinite. The sum $\mathfrak{F} + \mathfrak{F}(S_1) + \mathfrak{F}(S_2) + \mathfrak{F}(S_3)$ is a symmetric function of the quantities a, b, c, d ; one will be able to recognize if it is null without knowing the roots of the polynomial under the radical.

Examples

Let $R(t) = (1 - t^2)(1 - k^2 t^2)$. The integral

$$\int f(t) \frac{dt}{\sqrt{(1 - t^2)(1 - k^2 t^2)}}$$

will be pseudo-elliptic, provided that we have

$$f(t) + f(-t) + f\left(\frac{1}{kt}\right) + f\left(-\frac{1}{kt}\right) = 0.$$

The same will be true of the integral

$$\int \frac{f(x^2) dx}{\sqrt{(1 - x^2)(1 - k^2 x^2)}},$$

provided that one has

$$f(x^2) + f\left[\frac{1 - k^2 x^2}{k^2(1 - x^2)}\right] + f\left(\frac{1}{k^2 x^2}\right) + f\left(\frac{1 - x^2}{1 - k^2 x^2}\right) = 0.$$

(See HERMITE, Journal de Liouville, 1880; RAFFY, loc. cit., p. 71.)

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The preceding analysis shows us a relationship between some pseudo-elliptic integrals and the linear substitutions that permute the roots of the equation $R(t) = 0$. If the quantities a, b, c, d are arbitrary, there are no other linear substitutions than the three previous ones that permute these four roots. But, for some special forms of $R(t)$, there will be other substitutions of this kind. It is natural to wonder whether one can also attach pseudo-elliptic integrals to these new substitutions. This is indeed what happens.

Several assumptions can be made about these substitutions. Let a, b, c, d always be the roots of $R(t) = 0$; it may be that there is a linear substitution that does not change the point a and permutes the other three roots b, c, d circularly. This substitution will necessarily be of period 3.

Let S be such a substitution. Now let $F(t)$ be a rational function of t having the property expressed by the equation

$$F(S) = \alpha F \quad \text{where} \quad \alpha = e^{\frac{2i\pi}{3}},$$

the integral

$$\int F(t) \frac{dt}{\sqrt{R(t)}}$$

is pseudo-elliptic. Let's first do the transformation $z = \frac{t-a}{t-a_1}$, where a, a_1 are the fixed points of S . The transform of S by this substitution is $S'[z; \alpha z]$. The differential $\frac{dt}{\sqrt{R(t)}}$ turns into a differential of the form

$$\frac{dz}{\sqrt{z(Az^3 + B)}},$$

since the three non-zero roots of the new polynomial under the radical must be circularly permuted by the substitution S' . The rational function $F(t)$ changes into a rational function $F_1(z)$ verifying the relationship

$$F_1(\alpha z) = \alpha F_1(z),$$

which will, therefore, be in the form

$$F_1(z) = z\varphi(z^3).$$

The preceding differential becomes

$$\frac{\varphi(z^3)z dz}{\sqrt{z(Az^3 + B)}}.$$

It now suffices to set $z^3 = x$ to be reduced to a differential of the form (1); from which results the theorem stated above.

One can link to this case of reduction a rather famous integral, studied by Legendre and Clausen. If there is a linear substitution of period 3 leaving one of the roots a, b, c, d fixed and permuting the others circularly, one can always, by a linear substitution, bring $R(t)$ to the form $R(t) = t^3 - 1$; which amounts to supposing $a = \infty, b = 1, c = \alpha, d = \alpha^2$. There are eight period 3 substitutions, which permute three of these roots and leave the fourth fixed. To the four substitutions fixing one finite root correspond four types of pseudo-elliptic integrals. The substitutions to reduce these integrals are respectively

$$t^3 = x, \quad \left(\frac{t-1}{t+2}\right)^3 = x, \quad \left(\frac{t-\alpha}{t+2\alpha}\right)^3 = x, \quad \left(\frac{t-\alpha^2}{t+2\alpha^2}\right)^3 = x.$$

The first is immediate. As for the second, an easy calculation allows one to verify that by setting $x = \left(\frac{t-1}{t+2}\right)^3$, we have

$$\frac{1}{3} \frac{dx}{\sqrt{x(1-x)}} = \frac{t-1}{t+2} \frac{dt}{\sqrt{t^3-1}}.$$

Let A, B, C be three arbitrary constants; the integral

$$\int \left(A \frac{t-1}{t+2} + B \frac{t-\alpha^2}{t+2\alpha^2} + C \frac{t-\alpha}{t+2\alpha} \right) \frac{dt}{\sqrt{t^3-1}}$$

will be expressed in finite terms using elementary symbols. This integral can be written

$$\int \frac{\lambda(t^3-4) + \mu t^2 + \nu t}{t^3+8} \frac{dt}{\sqrt{t^3-1}}$$

where $\lambda = A + B + C$, $\mu = -3(A + B\alpha^2 + C\alpha)$, $\nu = 6(A + B\alpha + C\alpha^2)$. The coefficients λ, μ, ν are, like A, B, C , arbitrary. Therefore the integrals

$$\int \frac{(t^3-4)dt}{(t^3+8)\sqrt{t^3-1}}, \quad \int \frac{t^2 dt}{(t^3+8)\sqrt{t^3-1}}, \quad \int \frac{t dt}{(t^3+8)\sqrt{t^3-1}}$$

are pseudo-elliptic integrals; the last one is precisely the integral of Legendre and Clausen.

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The linear substitutions which permute the four quantities $\infty, 1, \alpha, \alpha^2$ form a group of twelve substitutions isomorphic to the group formed by the rotations which bring a regular tetrahedron onto itself. Let S_i be one of the twelve substitutions and μ_i the corresponding multiplier. Let us take a rational function $F(t)$ satisfying the two relations

$$\begin{cases} \sum_{i=1}^{12} F(S_i) = 0, \\ \sum_{i=1}^{12} \frac{1}{\mu_i} F(S_i) = 0; \end{cases} \quad (6)$$

the integral

$$\int F(t) \frac{dt}{\sqrt{t^3-1}}$$

is a pseudo-elliptic integral. To demonstrate this, I set

$$\begin{aligned} F(t) + F\left(\frac{t+2}{t-1}\right) + F\left(\frac{t+2\alpha}{\alpha^2 t-1}\right) + F\left(\frac{t+2\alpha^2}{\alpha t-1}\right) &= 4\varphi(t), \\ F(t) - F\left(\frac{t+2}{t-1}\right) + F\left(\frac{t+2\alpha}{\alpha^2 t-1}\right) - F\left(\frac{t+2\alpha^2}{\alpha t-1}\right) &= 4\varphi_1(t), \\ F(t) - F\left(\frac{t+2}{t-1}\right) - F\left(\frac{t+2\alpha}{\alpha^2 t-1}\right) + F\left(\frac{t+2\alpha^2}{\alpha t-1}\right) &= 4\varphi_2(t), \\ F(t) + F\left(\frac{t+2}{t-1}\right) - F\left(\frac{t+2\alpha}{\alpha^2 t-1}\right) - F\left(\frac{t+2\alpha^2}{\alpha t-1}\right) &= 4\varphi_3(t); \end{aligned}$$

from which we deduce

$$F(t) = \varphi(t) + \varphi_1(t) + \varphi_2(t) + \varphi_3(t).$$

But the functions $\varphi_1, \varphi_2, \varphi_3$ respectively satisfy the relations

$$\varphi_1\left(\frac{t+2}{t-1}\right) + \varphi_1(t) = 0, \quad \varphi_2\left(\frac{t+2\alpha}{\alpha^2 t-1}\right) + \varphi_2(t) = 0, \quad \varphi_3\left(\frac{t+2\alpha^2}{\alpha t-1}\right) + \varphi_3(t) = 0,$$

so that, according to the proposition of n°2, the three integrals

$$\int \varphi_1(t) \frac{dt}{\sqrt{t^3-1}}, \quad \int \varphi_2(t) \frac{dt}{\sqrt{t^3-1}}, \quad \int \varphi_3(t) \frac{dt}{\sqrt{t^3-1}}$$

are pseudo-elliptic integrals. We still have to consider the integral $\int \varphi(t) \frac{dt}{\sqrt{t^3-1}}$, but, taking into account the relations (6), we will have

$$\varphi(t) + \varphi(\alpha t) + \varphi(\alpha^2 t) = 0, \quad \varphi(t) + \alpha^2 \varphi(\alpha t) + \alpha \varphi(\alpha^2 t) = 0,$$

so that $\varphi(t)$ will be of the form $\varphi(t) = t^2 \psi(t^3)$, and the considered differential takes the form

$$t^2 \psi(t^3) \frac{dt}{\sqrt{t^3-1}},$$

which one reduces immediately to the form (1), by setting $t^3 = x$.

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One can still make two hypotheses on the linear substitutions which permute the four roots a, b, c, d :

1. It may be that a substitution, having for its double points two of these roots, permutes the other two.
2. It may happen that there exists a linear substitution, circularly permuting the four roots.

In the first case, the differential $\frac{dt}{\sqrt{R(t)}}$ can be reduced, by a linear substitution, to the form $\frac{dt}{\sqrt{t(t^2+1)}}$, and, in the second case, to the form $\frac{dt}{\sqrt{t^4-1}}$. Besides, one passes from the first to the second by changing t into $\frac{1-t}{1+t}$. It is therefore sufficient to consider the second form. There exist eight substitutions permuting the four quantities $+1, -1, +i, -i$; they are the following:

$$\begin{aligned} & [t; t], \quad [t; it], \quad [t; -t], \quad [t; -it], \\ & \left[t; \frac{1}{t} \right], \quad \left[t; \frac{i}{t} \right], \quad \left[t; -\frac{1}{t} \right], \quad \left[t; -\frac{i}{t} \right]; \end{aligned}$$

they form a group isomorphic to the group formed by the rotations which bring a system of two rectangular diameters of the sphere back upon itself. One can make correspond to these substitutions the following types of reducible integrals:

$$\int t \varphi(t^2) \frac{dt}{\sqrt{t^4-1}}, \quad \int (t \mp \frac{1}{t}) \varphi(t \pm \frac{1}{t}) \frac{dt}{\sqrt{t^4-1}}, \quad \dots$$

but it should be noted that the substitutions of period 4 do not provide new pseudo-elliptic integrals.

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Given a polynomial $R(t)$ of degree higher than the fourth, having only simple linear factors, there does not in general exist a linear substitution permuting the roots of the equation $R(t) = 0$. If such substitutions exist, one can make correspond to them ultra-elliptic integrals of the form

$$\int f[t, \sqrt{R(t)}] dt,$$

which it is possible to reduce, by an algebraic substitution, to integrals of a lower genus. Without entering into all the details here, I will recall the two following examples, which are well known.

If one is dealing with a sixth-degree polynomial whose roots are in involution, one knows that one can, by a linear substitution, reduce it to the form

$$R(u) = au^6 + bu^4 + cu^2 + d,$$

and the two integrals of the first and second kind

$$\int \frac{du}{\sqrt{R(u)}}, \quad \int \frac{u^2 du}{\sqrt{R(u)}}$$

are reduced by the substitution $u^2 = x$ to elliptic integrals. This is basically the case of reduction found by Jacobi (Crelle's Journal, t. 8).

As another example, let us take $R(t) = t(t^4 - 1)$; the six quantities $0, \infty, \pm 1, \pm i$ are permuted by a group of twenty-four linear substitutions, derived from the two substitutions

$$[t; it], \quad \left[t; \frac{1-t}{1+t} \right];$$

this group is isomorphic to the group formed by the rotations which bring a regular octahedron back upon itself. The substitutions of period 2, which permute the roots in pairs, are four in number:

$$\left[t; \frac{1-t}{1+t} \right], \quad \left[t; \frac{1+t}{1-t} \right], \quad \dots$$

To each of them correspond two integrals of the first kind, having only two distinct periods,

$$\int \frac{(t \pm 1 \pm \sqrt{2})dt}{\sqrt{t(t^4 - 1)}}, \quad \dots$$

According to a theorem due to M. Poincaré, one concludes from this that there will be an infinity of integrals of the form

$$\int \frac{(\alpha t + \beta)dt}{\sqrt{t(t^4 - 1)}}$$

having only two distinct periods.